Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

Sr.No. 8626

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block:409 Course Section:1	Seat No: 2022700003

1_32_616_986_1_1505_41402

Name of the candidate. Ashwika. Kailasan

Candidate's UID number: 2 0 2 2 7 0 0 0 0 3

Examination class room number: 4 0 9

Invigilator's signature with date: 08/10/25

(To be filled in by the candidate)

EXAMINATION: BE BTECH
(For example FY MCA/FY BTECH)

PROGRAM: CSE DS

DAY AND DATE: Wednesday, 8th Oct 2025

COURSE NAME: DL

SEMESTER : I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Space For Marks	Question No.	

Q1]

a) $f(\omega) = \frac{1}{2}(x\omega - y)^2$

$$\omega_{k+1} = \omega_k - \eta \frac{\partial f}{\partial \omega}$$

$$f = \frac{1}{2}(x^2\omega^2 - 2x\omega y + y^2)$$

$$\frac{\partial f}{\partial \omega} = \frac{1}{2}\left(2\omega x^2 - 2xy\right)$$

$$= \omega x^2 - xy$$
$$= x(\omega x - y)$$

eigen value for this is $\frac{xy}{x^2}$

$\frac{y}{x}$

$$\lambda_{max} = \frac{y}{x}$$

for convergence to ensure no divergence

$0 < \eta < \frac{\cancel{2}}{\lambda_{max}}$ ← when using Hessian matrix

$0 < \eta < \frac{y}{x}$

∴ when not hessian
↳ not x and y

∴ We have $\frac{2}{\lambda_{max}}$

We know.

$R = \frac{K-1}{K+1}$ where K is condition number

↓

rate

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here $\eta = \frac{\lambda_{max}}{\lambda_{min}}$

$\therefore R = \frac{\lambda_{max} - \lambda_{min}}{\lambda_{max} + \lambda_{min}}$

convergence rate

for $X^T X$

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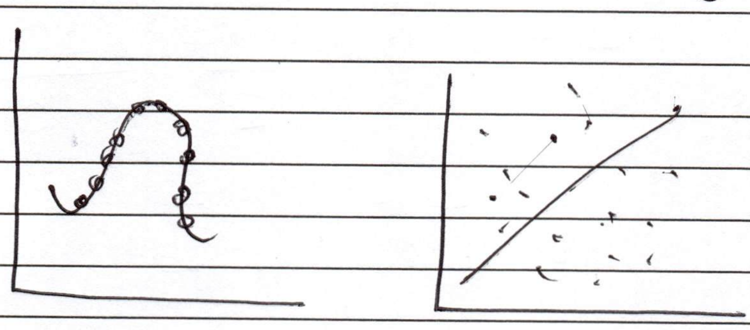
Q1]
b) $Loss = Bias^2 + Variance + k$

Underfitting → bias is high
overfitting → variance is high

our loss function is more affected by high bias ∴ squaring it.



high bias ↑ high variance.

Therefore model capacity being high can be a problem because this means the model can understand a wide variety of problems. It can predict correctly for a wide range of functions. This could mean model has large number of parameters and hyperparameters.

In cases of such models, the model may train with high accuracy because of above reasons, but could have problems with validation accuracy → which is the core issue of overfitting. This happens when model isn't able to apply its

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		prediction on new unseen data inspite of seeming great during training.

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Q2] a]

We know no. of params can be given by

$= (F_h \times F_w \times C_{in} - 1) \cdot C_{out}$

computational complexity of 2D conv is $O(k^2)$

Dept wise separable conv.

means the channels can be separated and treated as one and the combined later
∴ it is different from 2D convolution.

2D:-

on $(H \times W)$ apply $(F \times F)$

$\Rightarrow [(H-1) \times (W-1)] \times F \rightarrow$ total.

no. of multiplications.

assuming no padding and small k.

time complexity fo depth wise separable convolution will be.

$(H \times W) \times C \times K$

$\therefore O(k^3)$

↳ because we separate it out.

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	Q2] b)	

linear undercomplet AE trained with MSE → learns subspace equivalent to PCA.

This is because in an Undercomplete Autoencoder the size of the latentcode in bottle neck is smaller than that of input forcing it to learn important features only ∴ causing dimensionality reduction

Which is also done in case of PCA.

But wen. linear undercomplete autoencoder is trained with MSE

as in : $(\bar{O}-\bar{i})^2$

↳ it amplifies large errors and minimizes small error.
∴ highlighting only real critical errors.

this is also like approch of PCA where the x, y are first used to form covariance matrix using

$$\sum \frac{1}{n-1}(X_{1k}-\bar{X}_1)(X_{2k}-\bar{X}_2)$$

or

$$\sum \frac{1}{n-1}(X_{1k}-\bar{X}_1)^2$$

or 2.

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Which is similar to MSE
∴ ~~they can~~ proven.

an over complete autoencoder has a latent space code larger than the input ∴ problem can be faced where the model will just copy input as it is to output. due to no. of neurons > input. ~~has~~ learns all latent features as it is with no new modification.

This problem can be contained if we use regularizers like L1 and L2 regularization forming variants like:—

Sparse autoencoder:—

Shuts down excess neurons to avoid this problem.

Denoising autoencoder:—

Removes any corruption.

L1 and L2 ⟶

L1: the weights of neurons not helping much are made 0 ∴ completely eliminating them

L2: the weights of neurons are weighted according to how much they help not directly made 0 like L1.

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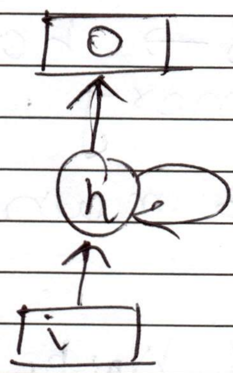
	Q3	
	a]	$\frac{\partial L}{\partial h_{t-k}}$.

RNN:-



eqn : $h_t = \tanh(W_{hh} h_{t-1}, W_{xh} x_t + b_i)$

$y_t = W_{hy} h_t + b$

for an RNN

loss can be givn by

$$L = \sum_{t=0}^{t=n} (y_{e_t} - y_{o_t})^2$$

now this is all affected by the hidden states which are reccursively altered.

$$\therefore h_{k+1} = h_k + \eta \frac{\partial L}{\partial h_{-1}}$$

η learning rate

instead of weights we have hidden states here, remaining is same.

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LSTMS solve the problem of vanishing memory faced by RNN ~~c~~ can't remember long sequences ∴ RNN is called short term memory

Wereas LSTMS → have long term memory.

They have 3 main cate

- Input ~~Input~~ $i = \varepsilon(W_i h_{t-1}, x_t + b_i)$
- Forget $f = \varepsilon(W_f h_{t-1}, x_t + b_f)$

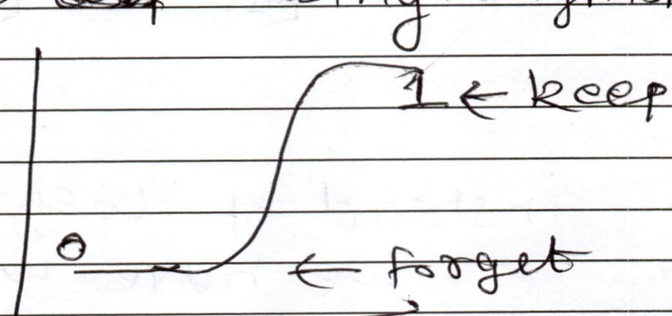
— tanh —

- Output $O = \varepsilon(W_o h_{t-1}, x_t + b_o)$

— tanh —

this solves problem because LSTMS do not unnecesarily store all memory, they only store relevant and discard the rest,

as we can see → forget gate applies the sigmoid funcino on input coming in from previous hidden states and decides whether to forget or keep ~~xxx~~ using sigmoid ←



∴ eliminating the problem by rembering long sequence.

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	Q3)	
	b)	GRU is alot like LSTM except it combines the input gate and forget gate into an update gate and also has a reset gate. It combines prev. hidden state with cell state ~~update gate~~
		update gate formula is alot like input / forget gate
		$u = \sigma(W_u h_{t-1}, x_t + b_u)$.

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GRUs have lesser parameters than LSTMs because of fewer number of gates. They have different time complexities with GRU being faster.

LSTMs cabe great incase of stock market analysis where the data needs to be that exact and accurate.

Whereas GRU can be use in an entirely differen approch like in overtime fMRI memorry analysis. Incase of mapping. Being mathematically more implementable → easily. It has the simplicity of an RNN as well as the advancements of an LSTM.

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